

PathBridger: Prefix-Calibrated State-Path Bridges for Offline Goal-Conditioned Reinforcement Learning

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Abstract

Offline goal-conditioned reinforcement learning (GCRL) must compose long-horizon behavior from fixed trajectories while avoiding unsupported rollouts and long open-loop execution. We propose *PathBridger*, a state-path proposal mechanism for offline GCRL. Given a current state and a commanded goal, PathBridger samples a local subgoal displacement, constructs a prefix-calibrated endpoint bridge in coordinates centered at the current state, and applies an endpoint-preserving residual correction to match dataset-supported path geometry. The resulting state path is reconstructed in the original state space, decoded into short action-prefix candidates by inverse dynamics, scored by chunked critics conditioned on the estimated subgoal, and distilled into a reactive actor through a single-step proximal policy improvement objective. The bridge is not used as a learned environment transition model. It serves as an endpoint-pinned proposal template whose progress is visible within the actor-executed prefix, while learning is allocated to subgoal selection, interior path correction, inverse dynamics, critic estimation, and actor extraction.

1 Introduction

Offline GCRL aims to learn policies that reach commanded goals using only a fixed trajectory dataset. Goal relabeling provides dense supervision from unannotated trajectories, but long-horizon domains remain difficult. Sparse rewards propagate slowly through one-step TD backups, learned values for distant goals often have low signal-to-noise ratios, and off-support actor updates can exploit unreliable value estimates. These issues are central in OGBench, which emphasizes stitching, long-horizon reasoning, and goal generalization in offline GCRL (Park et al. 2025b).

A natural way to reduce this difficulty is to shorten the effective decision horizon. Hierarchical GCRL decomposes faraway goals into nearer subgoals (Park et al. 2023). Action-chunk and chunked-critic methods accelerate value propagation by evaluating temporally extended action sequences rather than individual actions (Li, Zhou, and Levine 2025; Li, Park, and Levine 2025; Kim et al. 2025; Park et al. 2025a). These approaches show that longer temporal units can make offline RL more scalable. However, directly predicting and executing long action chunks can reduce reactivity, and modeling high-dimensional action sequences can be difficult in manipulation and navigation domains.

PathBridger takes a complementary state-space route. Instead of asking the actor to output a long open-loop action sequence, it proposes a short state path from the current state to a learned intermediate subgoal. Adjacent state pairs on the proposed path are decoded into short action-prefix candidates by inverse dynamics. Chunked critics score these candidates, and the actor is trained to make a local critic-improving update around the decoded prefixes. The actor therefore executes only a short prefix, while that prefix is informed by a longer subgoal-conditioned state proposal.

The central modeling choice is to construct paths in a translated displacement chart. Internally, every path starts at zero displacement and terminates at a sampled subgoal displacement. Externally, generated displacements are mapped back to the original state space by adding the current state. This chart separates local path shape from absolute state coordinates while still allowing the residual path corrector to condition on the current state. We further calibrate deterministic bridge progress to the actor horizon: because the actor executes only h_a steps of a K -step bridge, the reference path is scheduled so that the executed prefix already makes visible progress toward the selected subgoal.

The resulting contribution is a proposal-to-policy pipeline for offline GCRL. PathBridger learns *where* to bridge with a value-weighted Gaussian subgoal proposal, *how* to trace the bridge with an endpoint-preserving residual corrector, and *how* to execute it with inverse dynamics and proximal actor extraction. It deliberately does not learn the physical transition kernel. Instead, it learns a structured state-path proposal and a reactive actor around offline-supported short prefixes.

2 Related Work

Offline goal-conditioned reinforcement learning. Offline GCRL learns goal-reaching behavior from static trajectories, typically using goal relabeling and value functions. HIQL decomposes distant goal reaching through nearer subgoals and extracts hierarchical policies from a shared value function (Park et al. 2023). OGBench evaluates stitching, long-horizon reasoning, image observations, and stochasticity in offline GCRL (Park et al. 2025b). PathBridger shares the subgoal motivation, but it does not treat the subgoal as the only high-level action. It constructs a local state path to the subgoal and converts that path into executable action-prefix proposals.

Horizon reduction and action chunks. Recent work argues that horizon reduction is central to scaling offline RL (Park et al. 2025c). Q-Chunking learns values over action chunks (Li, Zhou, and Levine 2025); DQC separates the critic chunk length from the shorter policy execution horizon (Li, Park, and Levine 2025); DEAS uses action sequences with detached value learning (Kim et al. 2025); and MAC uses action-chunk models for model-based value expansion (Park et al. 2025a). PathBridger adopts the same horizon-reduction premise, but its temporally extended object is a state-space proposal rather than a policy-predicted action chunk. Short action prefixes are produced only after inverse-dynamics decoding and critic selection.

Diffusion bridges and stochastic path references. Diffusion bridge methods construct stochastic processes with fixed endpoints, often using Brownian, Ornstein–Uhlenbeck, or stochastic optimal-control references. GOUB applies a generalized Ornstein–Uhlenbeck bridge to image restoration (Yue et al. 2024), while UniDB interprets diffusion bridges through stochastic optimal control and relaxes hard endpoint enforcement through a terminal penalty (Zhu et al. 2025). PathBridger uses this viewpoint only as an analytic endpoint-conditioned reference for state-path proposals. Unlike image-restoration bridges, it is not trained as a full generative diffusion model and is not intended to approximate the environment transition kernel.

Offline actor improvement. Offline actor updates must improve value while remaining near actions for which the critic is reliable. IQL uses implicit value learning and advantage-weighted extraction (Kostrikov, Nair, and Levine 2022), whereas TD3+BC adds an explicit behavior-cloning anchor to actor improvement (Fujimoto and Gu 2021). PathBridger uses a related proximal principle: bridge-induced inverse-dynamics candidates define an empirical reference distribution, and the actor performs a single critic-improving proximal step around that distribution.

3 Background

This section fixes notation and summarizes the three ingredients used by PathBridger: decoupled chunked critics for horizon reduction, transitive state values for subgoal selection, and endpoint bridges for state-path proposals.

3.1 Offline GCRL Notation

We consider an offline goal-conditioned MDP with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel p , and a fixed dataset $\mathcal{D}$ of trajectories

$$\tau = (s_0, a_0, s_1, a_1, \dots).$$

At evaluation time, the agent receives a current state $s_t \in \mathbb{R}^d$ and a final goal s_g . We write

$$z_g = \phi(s_g)$$

for the compact goal representation used by the subgoal proposal, critics, and value function. The goal-conditioned value function is denoted by $V_\xi(s, z_g)$.

PathBridger uses three horizons. The state-path bridge has horizon K , the actor executes a short prefix of length $h_a \leq$

K , and the chunked critic may use a longer horizon H_Q . For an offline segment

$$(s_t^{\mathcal{D}}, s_{t+1}^{\mathcal{D}}, \dots, s_{t+K}^{\mathcal{D}}),$$

we define the local displacement chart

$$r_i^{\mathcal{D}} = s_{t+i}^{\mathcal{D}} - s_t^{\mathcal{D}}, \quad r_0^{\mathcal{D}} = 0, \quad \delta^{\mathcal{D}} = r_K^{\mathcal{D}}. \quad (1)$$

At inference, generated displacements are reconstructed by

$$\hat{s}_{t+i} = s_t + \hat{r}_i. \quad (2)$$

Thus bridge construction and path supervision occur in local r -coordinates, while inverse dynamics, critics, and the actor consume reconstructed states.

3.2 Decoupled Q-Chunking

We use DQC as the value-learning template. A chunked critic estimates the value of an action sequence, improving credit assignment in long-horizon tasks. Asking the policy to predict the same long chunk, however, can be difficult and overly open-loop. DQC therefore decouples the critic horizon from the policy horizon: a full critic evaluates long chunks, while a partial critic evaluates the shorter prefix executed by the policy (Li, Park, and Levine 2025).

In our goal-conditioned notation, a full chunk critic has the form

$$Q_\varphi^H(s_t, a_{t:t+H_Q-1}, z_g), \quad (3)$$

and may be trained with a multi-step target

$$\mathcal{L}_{Q^H} = \mathbb{E} \left[(Q_\varphi^H(s_t, a_{t:t+H_Q-1}, z_g) - y_t^H)^2 \right], \quad (4)$$

where

$$y_t^H = \sum_{j=0}^{H_Q-1} \gamma^j r(s_{t+j}, z_g) + \gamma^{H_Q} \bar{V}_\xi(s_{t+H_Q}, z_g). \quad (5)$$

A partial critic is trained as an optimistic projection of the full critic onto a shorter action prefix:

$$\mathcal{L}_{Q^P} = \mathbb{E} \left[\ell_{\kappa_P} \left(\bar{Q}_\varphi^H(s_t, a_{t:t+H_Q-1}, z_g) - Q_\varphi^P(s_t, a_{t:t+h_a-1}, z_g) \right) \right]. \quad (6)$$

where ℓ_{κ_P} is an asymmetric expectile or quantile loss. The value function is trained by implicit maximization over partial chunks,

$$\mathcal{L}_V = \mathbb{E} \left[\ell_{\kappa_V} \left(\bar{Q}_\varphi^P(s_t, a_{t:t+h_a-1}, z_g) - V_\xi(s_t, z_g) \right) \right]. \quad (7)$$

PathBridger adopts this decoupled value-learning viewpoint, but obtains candidate partial chunks from state-path bridges and inverse dynamics rather than from an action-chunk policy.

3.3 Transitive State Values

Following Transitive RL (TRL) (Park et al. 2026), we use the triangle-inequality structure of goal-conditioned reachability to define a divide-and-conquer state-value objective. For the subgoal objective, it is useful to distinguish the state-space value used to select intermediate waypoints from the action-chunk critic used for actor improvement. We introduce a state-pair transitive value

$$V_\eta(s, g) \in (0, 1), \quad (8)$$

which estimates the discounted reachability of a state goal g from state s . In the ideal dynamic-programming limit, this value satisfies the transitive composition rule

$$V_\eta^*(s, g) = \max_{z \in S} V_\eta^*(s, z) V_\eta^*(z, g). \quad (9)$$

Thus an optimal intermediate state maximizes the product of the value of reaching the intermediate state and the value of continuing from that state to the final goal. Equivalently, if $V_\eta(s, g) = \gamma^{D_\eta(s, g)}$ for a temporal distance D_η , then Eq. (9) becomes the min-plus shortest-path relation

$$D_\eta^*(s, g) = \min_{z \in S} [D_\eta^*(s, z) + D_\eta^*(z, g)]. \quad (10)$$

As in TRL, offline learning replaces the maximization over arbitrary states with in-trajectory intermediate states and an asymmetric regression loss, avoiding unconstrained maximization over unsupported subgoals. For a sampled trajectory triple $i < k < j$, define

$$y_{ij}^{\text{tr}} = \text{sg}[\bar{V}_\eta(s_i, s_k) \bar{V}_\eta(s_k, s_j)]. \quad (11)$$

The transitive value loss is

$$\mathcal{L}_V^{\text{tr}} = \mathbb{E}_{i < k < j} [\ell_{\tau_V}(y_{ij}^{\text{tr}} - V_\eta(s_i, s_j))], \quad (12)$$

where $\tau_V > 1/2$ makes the regression an in-support soft maximum over sampled splits. A short-range base loss anchors the scale:

$$\mathcal{L}_V^{\text{base}} = \mathbb{E}_{1 \leq d \leq h_b} [(V_\eta(s_i, s_{i+d}) - \gamma^d)^2]. \quad (13)$$

The state-transitive value objective is

$$\mathcal{L}_V^{\text{state}} = \mathcal{L}_V^{\text{base}} + \lambda_{\text{tr}} \mathcal{L}_V^{\text{tr}}. \quad (14)$$

This value is not obtained as a readout from an action-chunk critic. It is a primary state-space potential used to evaluate whether a proposed subgoal lies on a high-value route to the final goal.

3.4 Linear-SDE Endpoint Bridges

PathBridger uses an endpoint bridge as an analytic state-path reference. Consider the passive linear-SDE process in displacement coordinates,

$$dr_\tau = \theta_\tau r_\tau d\tau + g_\tau dW_\tau, \quad g_\tau^2 = 2\lambda^2 \theta_\tau. \quad (15)$$

For a discrete interval $i \rightarrow i+1$, define

$$A_i = e^{\theta_i}, \quad q_i^2 = \lambda^2(e^{2\theta_i} - 1). \quad (16)$$

For $0 \leq i \leq j \leq K$, write

$$\Theta_{i:j} = \sum_{\ell=i}^{j-1} \theta_\ell, \quad F_{i:j} = e^{\Theta_{i:j}}, \quad S_{i:j} = \lambda^2(F_{i:j}^2 - 1). \quad (17)$$

Then

$$r_j \mid r_i \sim \mathcal{N}(F_{i:j} r_i, S_{i:j} I). \quad (18)$$

Imposing a terminal displacement through

$$\delta = r_K + \zeta, \quad \zeta \sim \mathcal{N}(0, \varepsilon_\gamma I), \quad (19)$$

yields the bridge marginal

$$r_i \mid \delta \sim \mathcal{N}(\beta_i^\gamma \delta, B_i^\gamma I), \quad (20)$$

where

$$\beta_i^\gamma = \frac{S_{0:i} F_{i:K}}{S_{0:K} + \varepsilon_\gamma}, \quad B_i^\gamma = \frac{S_{0:i}(S_{i:K} + \varepsilon_\gamma)}{S_{0:K} + \varepsilon_\gamma}. \quad (21)$$

The hard bridge corresponds to $\varepsilon_\gamma = 0$. In PathBridger, this bridge is a pinned proposal reference rather than a learned transition model.

4 PathBridger

PathBridger answers three linked questions. First, *which local subgoal should the agent aim for?* A Gaussian subgoal-displacement proposal selects a dataset-supported waypoint useful for the final goal. Within the bridge dynamics, this subgoal is the terminal endpoint of the local bridge. Second, *what state path should connect the current state to this subgoal?* A prefix-calibrated bridge gives a pinned reference path, and a residual corrector bends its interior toward dataset geometry. Third, *which executable action prefix should the actor take?* Inverse dynamics decodes candidate prefixes from generated paths, chunked critics score them conditioned on the estimated subgoal, and the actor performs a proximal improvement step around the critic-weighted proposal distribution.

4.1 Choosing a Local Subgoal

The first component selects a local subgoal for the agent. In the bridge construction, this subgoal is the terminal endpoint of the local bridge. Rather than predicting an absolute future state directly, PathBridger models the subgoal displacement

$$\delta = s_{t+K} - s_t.$$

The subgoal proposal is a diagonal Gaussian,

$$G_\chi^r(\delta \mid s_t, z_g) = \mathcal{N}(\mu_\chi^r(s_t, z_g), \text{diag}((\sigma_\chi^r)^2(s_t, z_g))). \quad (22)$$

A sample is drawn by reparameterization,

$$\hat{\delta} = \mu_\chi^r(s_t, z_g) + \sigma_\chi^r(s_t, z_g) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (23)$$

and converted to an absolute subgoal by

$$\hat{s}_{\text{sg}} = s_t + \hat{\delta}. \quad (24)$$

Equivalently, $G_\chi(\hat{s}_{\text{sg}} \mid s_t, z_g) = G_\chi^r(\hat{s}_{\text{sg}} - s_t \mid s_t, z_g)$.

The subgoal proposal should remain close to data, but not all dataset future states are equally useful for the commanded goal. We therefore weight the likelihood of a dataset displacement by its estimated value improvement,

$$\Delta V_t = V_\xi(s_{t+K}^{\mathcal{D}}, z_g) - V_\xi(s_t, z_g), \quad (25)$$

using a clipped exponential weight

$$w_t = \text{sg}[\exp(\text{clip}(c_v \Delta V_t, \ell_{\min}, \ell_{\max}))]. \quad (26)$$

The clipping prevents unstable likelihood weights from dominating training. The subgoal loss is

$$\mathcal{L}_{\text{sg}} = \mathbb{E} \left[w_t \left(-\log G_\chi^r(\delta^{\mathcal{D}} \mid s_t, z_g) \right) - \alpha_{\text{sg}} \bar{V}_\xi(s_t + \hat{\delta}, z_g) \right]. \quad (27)$$

where $\hat{\delta} \sim G_\chi^r(\cdot \mid s_t, z_g)$. The first term fits dataset-supported K -step subgoal displacements, while the second term encourages sampled subgoals that are valuable for reaching the final goal. The value network is fixed inside the second term.

Transitive value-guided subgoal bonus. When the state-transitive critic is enabled, the value bonus in Eq. (27) is replaced by a transitive waypoint score. For a sampled subgoal $\hat{s}_{\text{sg}} = s_t + \hat{\delta}$, define

$$P_{\text{tr}}(s_t, \hat{s}_{\text{sg}}, s_g) = \bar{V}_\eta(s_t, \hat{s}_{\text{sg}}) \bar{V}_\eta(\hat{s}_{\text{sg}}, s_g). \quad (28)$$

A scale-normalized version compares the route through $\hat{s}_{\text{sg}}$ against the direct state-pair value:

$$B_{\text{tr}}(s_t, \hat{s}_{\text{sg}}, s_g) = \frac{\bar{V}_\eta(s_t, \hat{s}_{\text{sg}}) \bar{V}_\eta(\hat{s}_{\text{sg}}, s_g)}{\bar{V}_\eta(s_t, s_g) + \epsilon_V}. \quad (29)$$

The transitive subgoal objective is

$$\mathcal{L}_{\text{sg}}^{\text{tr}} = \mathbb{E} \left[-\log G_\chi^r(\delta^{\mathcal{D}} \mid s_t, z_g) - \alpha_{\text{tr}} B_{\text{tr}}(s_t, s_t + \hat{\delta}, s_g) \right], \quad \hat{\delta} \sim G_\chi^r(\cdot \mid s_t, z_g). \quad (30)$$

The likelihood term remains the supervised anchor to the dataset K -step subgoal displacement, while the transitive bonus tilts the density toward sampled subgoals that preserve the compositional value $V(s_t, \hat{s}_{\text{sg}})V(\hat{s}_{\text{sg}}, s_g)$. The value-network parameters are treated as fixed in this term, but gradients are allowed to flow through the sampled subgoal.

4.2 Drawing a Prefix-Calibrated State Path

After choosing a subgoal, PathBridger proposes an interior state path. The chosen subgoal acts as the terminal endpoint of the bridge in displacement coordinates. The core path parameterization is

$$\tilde{r}_i = c_i \delta + \eta_i R_\psi(s_t, \delta, i/K), \quad i = 0, \dots, K. \quad (31)$$

This is the main state-path equation of PathBridger. The first term gives a prefix-calibrated endpoint bridge from zero displacement to the subgoal displacement δ . The second term is an interior path-shape correction. The residual network also

receives the absolute anchor s_t , so the proposal remains local in displacement coordinates while its shape can depend on global context such as maze position or object configuration.

The main text only requires the schedules to satisfy the endpoint constraints

$$0 = c_0 \leq c_1 \leq \dots \leq c_K = 1, \quad \eta_0 = \eta_K = 0, \quad \eta_i \geq 0. \quad (32)$$

These conditions imply

$$\tilde{r}_0 = 0, \quad \tilde{r}_K = \delta,$$

so the learned residual cannot move either endpoint. In our experiments, c_i is chosen to make progress visible in the actor-executed prefix, while η_i is a simple endpoint-vanishing envelope. The concrete schedules are listed in Appendix A.

If stochastic candidates are enabled, we sample around the corrected path using an endpoint-vanishing bridge standard deviation b_i :

$$\hat{r}_i = \tilde{r}_i + \sigma_{\text{plan}} b_i \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, I), \quad (33)$$

and then clamp $\hat{r}_0 = 0$ and $\hat{r}_K = \delta$. When $\sigma_{\text{plan}} = 0$, the proposal is deterministic and $\hat{r}_i = \tilde{r}_i$. The standard deviation schedule b_i is derived from the analytic bridge variance in Appendix A; it is not a separate learned dynamics model.

The path correction is trained in displacement coordinates. For a dataset segment, compute $r_i^{\mathcal{D}}$ and $\delta^{\mathcal{D}}$ as in Eq. (1), and set $\delta = \delta^{\mathcal{D}}$. The path loss is

$$\mathcal{L}_{\text{path}} = \mathbb{E} \left[\frac{1}{\Lambda_K} \sum_{i=1}^{K-1} \lambda_i \|\tilde{r}_i - r_i^{\mathcal{D}}\|_1 \right]. \quad (34)$$

where $\lambda_i \geq 0$ and $\Lambda_K = \sum_{i=1}^{K-1} \lambda_i$. Uniform weights recover the standard interior path-matching loss, while larger weights on early indices emphasize the actor-visible prefix. This loss teaches the residual to correct the bridge interior toward offline path geometry while preserving the endpoint constraints in Eq. (32).

4.3 Action-Prefix Proposals

A proposed state path is useful only if it can be executed. PathBridger converts each reconstructed path into candidate action prefixes with inverse dynamics. For candidate path m ,

$$\hat{s}_{t+i}^{(m)} = s_t + \hat{r}_i^{(m)},$$

we decode

$$\hat{a}_i^{(m)} = I_\omega \left(\hat{s}_{t+i}^{(m)}, \hat{s}_{t+i+1}^{(m)} \right), \quad i = 0, \dots, h_a - 1. \quad (35)$$

Let $\hat{a}^{(m)} := \hat{a}_{0:h_a-1}^{(m)}$. The inverse-dynamics model is trained by one-step supervision,

$$\mathcal{L}_{\text{idm}} = \mathbb{E}_{(s_t, a_t, s_{t+1}) \sim \mathcal{D}} \left[\|I_\omega(s_t, s_{t+1}) - a_t\|_2^2 \right]. \quad (36)$$

Let $\hat{s}_{\text{sg}}^{(m)}$ denote the estimated subgoal that generated candidate m , and let $\hat{z}_{\text{sg}}^{(m)} = \phi(\hat{s}_{\text{sg}}^{(m)})$. The decoded prefixes are scored by the partial critic,

$$q_m = \bar{Q}_\varphi^P \left(s_t, \hat{a}^{(m)}, \hat{z}_{\text{sg}}^{(m)} \right), \quad (37)$$

and converted into Q-Boltzmann weights,

$$\rho_m = \frac{\exp(\beta_{\text{sbi}} \text{sg}(q_m))}{\sum_{j=1}^M \exp(\beta_{\text{sbi}} \text{sg}(q_j))}. \quad (38)$$

This defines an empirical proposal distribution over action prefixes,

$$\pi_{\text{invdyn}}^Q(\cdot | s_t, \{\hat{s}_{\text{sg}}^{(m)}\}_{m=1}^M) = \sum_{m=1}^M \text{sg}(\rho_m) \delta_{\text{sg}(\hat{a}^{(m)})}(\cdot). \quad (39)$$

Stop-gradient operators make the actor treat subgoal sampling, bridge construction, inverse-dynamics decoding, and proposal scores as fixed references. The subgoal associated with the selected proposal is forwarded to the actor update.

4.4 Extracting a Short-Prefix Actor

The actor is trained by a single proximal policy improvement step around the inverse-dynamics proposal. For fixed $(s_t, \hat{s}_{\text{sg}})$, let $\hat{z}_{\text{sg}} = \phi(\hat{s}_{\text{sg}})$ and define the critic-induced energy of a policy distribution π as

$$\mathcal{E}_{s_t, \hat{s}_{\text{sg}}}(\pi) = -\mathbb{E}_{a \sim \pi} [\bar{Q}_\varphi^P(s_t, a, \hat{z}_{\text{sg}})]. \quad (40)$$

A Wasserstein single-step proximal update from π_{invdyn}^Q is

$$\pi^+ \in \arg \min_{\pi} \left\{ \mathcal{E}_{s_t, \hat{s}_{\text{sg}}}(\pi) + \frac{1}{2\tau_{\text{sbi}}} W_2^2(\pi, \pi_{\text{invdyn}}^Q) \right\}. \quad (41)$$

The first term improves the actor under the partial critic conditioned on the estimated subgoal, while the second term keeps the update close to bridge-induced, inverse-dynamics-decoded prefixes.

In implementation, the actor is deterministic,

$$\pi_\phi(\cdot | s_t, \hat{s}_{\text{sg}}) = \delta_{\mu_\phi(s_t, \hat{s}_{\text{sg}})}. \quad (42)$$

For a Dirac actor and an empirical proposal distribution,

$$W_2^2(\delta_\mu, \pi_{\text{invdyn}}^Q) = \sum_{m=1}^M \text{sg}(\rho_m) \left\| \mu - \text{sg}(\hat{a}^{(m)}) \right\|_2^2. \quad (43)$$

Substituting Eq. (43) into Eq. (41) yields the following loss, where $\mu_t = \mu_\phi(s_t, \hat{s}_{\text{sg}})$:

$$\mathcal{L}_{\text{SPI}} = \mathbb{E} \left[-\bar{Q}_\varphi^P(s_t, \mu_t, \hat{z}_{\text{sg}}) + \frac{1}{2\tau_{\text{sbi}}} \sum_{m=1}^M \text{sg}(\rho_m) \left\| \mu_t - \text{sg}(\hat{a}^{(m)}) \right\|_2^2 \right]. \quad (44)$$

The temperature β_{sbi} controls how sharply the proposal concentrates on high-value candidates, while τ_{sbi} controls the strength of the proximal anchor.

4.5 Critic Learning and Joint Update

The critics give meaning to both subgoal selection and actor extraction. We use the DQC-style full critic, partial critic, and value losses introduced above. The critic goal argument

is a generic goal representation z : during critic training it is supplied by relabeled dataset goals, while during proposal scoring and SPI it is instantiated by the estimated subgoal representation $\hat{z}_{\text{sg}} = \phi(\hat{s}_{\text{sg}})$. The partial critic has the form

$$Q_\varphi^P(s_t, a_{t:t+h_a-1}, z). \quad (45)$$

Accordingly, the partial critic and value losses used in Path-Bridger are

$$\mathcal{L}_{Q^P} = \mathbb{E} \left[\ell_{\kappa_P} \left(\bar{Q}_\varphi^H(s_t, a_{t:t+H_Q-1}, z) - Q_\varphi^P(s_t, a_{t:t+h_a-1}, z) \right) \right]. \quad (46)$$

$$\mathcal{L}_V = \mathbb{E} \left[\ell_{\kappa_V} \left(\bar{Q}_\varphi^P(s_t, a_{t:t+h_a-1}, z) - V_\xi(s_t, z) \right) \right]. \quad (47)$$

State-transitive value and local chunk critic. The transitive variant separates the value used for subgoal learning from the action-chunk critic used for SPI. The state value $V_\eta(s, g)$ is trained by Eqs. (12)–(14); it is not an implicit readout of the action critic. The action critic is instead a local realization critic

$$Q_\theta^{\text{loc}}(s_t, a_{t:t+h_a-1}, z), \quad (48)$$

which evaluates whether a short action prefix can realize or make progress toward the selected subgoal z . For an offline action prefix ending at s_{t+h_a} , its bootstrapped target is

$$y_t^{\text{loc}}(z) = \gamma^{h_a} \bar{V}_\eta(s_{t+h_a}, z), \quad (49)$$

with the usual terminal replacement $y_t^{\text{loc}}(s_{t+d}) = \gamma^d$ when the queried goal is reached within the prefix. The local chunk critic loss is

$$\mathcal{L}_{Q^{\text{loc}}} = \mathbb{E} [\ell_Q(\text{sg}(y_t^{\text{loc}}(z)) - Q_\theta^{\text{loc}}(s_t, a_{t:t+h_a-1}, z))]. \quad (50)$$

Thus the dependency is $V \rightarrow Q^{\text{loc}}$, not $Q^{\text{loc}} \rightarrow V$.

For proposal scoring, candidate m has a subgoal endpoint $\hat{s}_{\text{sg}}^{(m)}$ and decoded action prefix $\hat{a}^{(m)}$. We combine a local action-feasibility score and a global transitive subgoal score,

$$q_m^{\text{tr}} = \lambda_Q \bar{Q}_\theta^{\text{loc}}(s_t, \hat{a}^{(m)}, \hat{s}_{\text{sg}}^{(m)}) + \lambda_V B_{\text{tr}}(s_t, \hat{s}_{\text{sg}}^{(m)}, s_g). \quad (51)$$

The Q-Boltzmann weights in Eq. (38) can then use q_m^{tr} in place of q_m . In this configuration, V_η decides which subgoals are globally useful, while Q_θ^{loc} decides which decoded action prefixes are locally executable.

All components are trained concurrently with

$$\begin{aligned} \mathcal{L}_{\text{joint}} = & w_{\text{sg}} \mathcal{L}_{\text{sg}} + w_{\text{path}} \mathcal{L}_{\text{path}} + w_{\text{idm}} \mathcal{L}_{\text{idm}} \\ & + w_H \mathcal{L}_{Q^H} + w_P \mathcal{L}_{Q^P} \\ & + w_V \mathcal{L}_V + w_\pi \mathcal{L}_{\text{SPI}}. \end{aligned} \quad (52)$$

This objective should be read functionally rather than as a list of unrelated losses: the subgoal loss learns where to bridge, the path loss learns how to trace the bridge through dataset geometry, inverse dynamics makes the path executable, and the critic/SPI losses decide which decoded prefixes should be distilled into the actor.

Algorithm 1 PathBridger training update

- 1: Sample trajectory segments from the offline dataset.
 - 2: Compute displacements $r_i^{\mathcal{D}}$ and subgoal displacement $\delta^{\mathcal{D}}$.
 - 3: Fit the subgoal proposal using Eq. (27).
 - 4: Fit the residual path corrector using Eq. (34).
 - 5: Fit inverse dynamics using Eq. (36).
 - 6: Update full critic, partial critic, and value using Eqs. (4), (46), and (47).
 - 7: If the transitive variant is enabled, update V_η and Q_θ^{loc} using Eqs. (14) and (50).
 - 8: Sample subgoal displacements and generate bridge paths using Eqs. (23)–(33).
 - 9: Decode action-prefix candidates by inverse dynamics and score them with Eq. (37).
 - 10: Form Q-Boltzmann proposal weights using Eq. (38).
 - 11: Update the actor using the SPI loss in Eq. (44).
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PathBridger therefore learns G_χ^r , R_ψ , I_ω , Q^H , Q^P , V , and μ_ϕ . The progress schedule c_i , the residual gate η_i , and the optional bridge standard deviation b_i provide a structured state-path template with pinned bridge endpoints and prefix-aware progress. They are fixed proposal schedules, not learned physical dynamics.

A Scheduling Details

This appendix lists the concrete schedules used in PathBridger. The main path parameterization in Eq. (31) only requires the endpoint conditions in Eq. (32).

Progress schedule. The default progress schedule is

$$c_i = \left(\frac{i}{K}\right)^\alpha, \quad \alpha \leq 1. \quad (53)$$

The exponent α controls how much progress is visible in the actor-executed prefix. For example, with $K = 25$, $h_a = 5$, and $\alpha = 0.8$, the prefix reaches

$$c_{h_a} = \left(\frac{5}{25}\right)^{0.8} \approx 0.276$$

of the terminal displacement.

Bridge-clock realization. The same progress schedule can be realized as the hard marginal mean of a linear-SDE endpoint bridge. Let

$$\Theta_i = \text{asinh}(c_i \sinh(\Theta_{\text{tot}})), \quad \theta_i = \Theta_{i+1} - \Theta_i. \quad (54)$$

For the hard bridge, the mean coefficient becomes

$$\beta_i = \frac{\sinh(\theta_i)}{\sinh(\Theta_{\text{tot}})} = c_i. \quad (55)$$

Thus $c_i \delta$ in Eq. (31) is the hard-bridge marginal mean under this clock.

Residual gate. The default residual gate is the quadratic endpoint envelope

$$\eta_i = \frac{i(K-i)}{K^2}. \quad (56)$$

It is a simple boundary-condition device: it vanishes at both endpoints and is positive in the bridge interior, so R_ψ acts as a path-shape correction rather than an endpoint mover.

Optional stochastic envelope. For stochastic candidates, the hard-bridge variance defines the standard deviation. With $\Theta_K = \Theta_{\text{tot}}$,

$$B_i = \lambda^2 \frac{(e^{2\Theta_i} - 1)(e^{2(\Theta_{\text{tot}} - \Theta_i)} - 1)}{e^{2\Theta_{\text{tot}}} - 1}, \quad b_i = \sqrt{B_i}. \quad (57)$$

Then $b_0 = b_K = 0$, so sampled perturbations in Eq. (33) respect the bridge endpoints after endpoint clamping.

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